

Production functional realisation – source audit and reference closure

TECT verification-first repository · claim A1-PRODUCTION-FUNCTIONAL-REALISATION

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1. Purpose and scope

This revision retains the hash-pinned external-source audit and adds a separate, explicit reference functional. The reference is an all-coupling mathematical candidate, not an assertion that the external working branch has been repaired. Neither result enlarges the pure-Brazovskii scalar-slice T5 claim.

2. Reference functional

On the stated real torus pairing, define $J_A = \nabla m_A$, $K_A = J_A - (m_A/(\rho + 10^{-12}))\nabla\rho$, and

$$F_{\text{ref}} = F_{\text{Braz}} + F_{\text{family}} + F_{\text{lock}} + F_{\text{shell}} + \int \left(\frac{\lambda}{4} \rho^2 + \frac{\gamma}{6} \rho^3 \right) + \sum_A \int \left(\frac{a}{2} |J_A|^2 + b \operatorname{Re}(\bar{J}_A \cdot K_A) + \frac{c}{2} |K_A|^2 \right),$$

where $a = c_{JJ}\alpha_X^2/(M_X^2 + 10^{-12})$, $b = c_{JK}\alpha_X\beta_X/(M_X^2 + 10^{-12})$, and $c = c_{KK}\beta_X^2/(M_X^2 + 10^{-12})$. Thus all three Class-II coefficients are represented and the scalar coefficients match the frozen real gradient.

3. Procedure and result

Torch autodiff supplies R_{ref} from F_{ref} . A centred derivative of that gradient supplies H_{ref} because this CPU runtime does not expose a reliable complex-FFT second-autodiff path. Independent energy finite differences and three outer derivative steps test the three identities over zero, homogeneous, random, commensurate q_0 -shell, and Class-II-active fields, both at the pinned point and a manifest-declared shell-bias activation.

The persisted reference run passes: maximum errors are 1.174×10^{-7} for $DF_{\text{ref}} = \langle R_{\text{ref}}, v \rangle_{\mathbb{R}}$, 1.005×10^{-9} for $DR_{\text{ref}}v = H_{\text{ref}}v$, and 3.380×10^{-11} for real Hessian symmetry. The independent source audit continues to fail variational closure, as required by its separate falsifier.

4. Devil’s-advocate

(alpha) “*The reference silently drops the troublesome Class-II term.*” DISMISSED: the manifest requires nonzero c_{JJ}, c_{JK}, c_{KK} and the cross term is present explicitly.

(beta) “*Autodiff makes the variational identity tautological.*” VALID with mitigation: the directional derivative is checked by independent energy finite differences at three steps, and $DR = H$ is checked at separate outer steps.

(gamma) “*Reference closure repairs production.*” UPHELD as an invalid reading: the external backend hash remains pinned and its audit remains failed; implementation authority is still an open engineering and review action.

5. Result footer

Result ID:	A1-PRODUCTION-FUNCTIONAL-REALISATION
Precise statement:	Proposed all-coupling reference functional closes the three discrete variational identities on N=4 tests.
Scope:	Proposed reference only; source audit remains separate.
Dependencies:	A1-PRODUCTION-KERNEL-MANIFEST.
Evidence grade:	EXECUTED, CONDITIONAL.
Reproduction command:	python codes/foundations/a1_production_functional_realisation.py --reference-closure
Expected output:	AUDIT-PASS-MISMATCH-DETECTED and REFERENCE-CLOSURE-PASS.
Falsification gate:	a failed field/step/identity or source/reference conflation.
Tier before / after:	T3 / T3 (no promotion).
No-overclaim statement:	Not a repaired production backend, PDE theorem, or BCC result.
Next required action:	implement the chosen reference convention in a separately hash-pinned production backend and rerun this audit.